FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

FIITJEE

JEE(Advanced)-2015

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
CRT–III
(Paper-1)

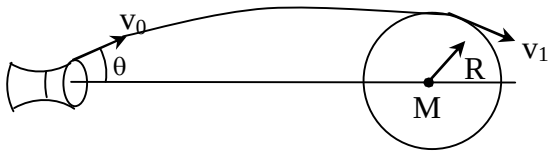
Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	C	B	C
2.	D	C	D
3.	D	C	A
4.	D	C	B
5.	B	D	A
6.	C	D	C
7.	B	C	C
8.	B	A	B
9.	A, C, D	B, C, D	A, C
10.	A, D	A, B, C	A, B, C
11.	A, B, D	A, B, D	A, B
12.	A, B, D	A, B, C	A, C
13.	C	B	A
14.	A	B	D
15.	A	D	A
16.	D	C	C
17.	A	A	B
18.	C	B	B
1.	4	2	0
2.	8	1	7
3.	1	3	3
4.	4	4	2
5.	8	2	2

Physics

PART – I

SECTION – A

1. Angular momentum conservation about the centre of the planet,



$$mv_0 \cdot 5R\sin\theta = mv_1R$$

$$v_1 = 5\,v_0\sin\theta \text{ -------(1)}$$

Energy conservation gives,

$$\frac{1}{2}mv_0^2 - \frac{GMm}{5R} = \frac{1}{2}mv_1^2 - \frac{GMm}{R} \text{ ------(2)}$$

Solve equation (1) and (2),

$\theta = 37°$

2. Assume a Gaussian surface to be a sphere of radius r and take limit as $r \to \infty$. The flux,

$$\int \vec{E}.d\vec{A} = \frac{q}{\varepsilon_0}$$

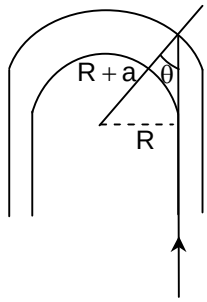
$$\int E.dA = \frac{q}{\varepsilon_0}$$

$$\lim_{r\to\infty} \frac{a}{r^2}e^{-r/b}\cdot 4\pi r^2 = \frac{q}{\varepsilon_0}$$

$$4\pi a \cdot \lim_{r\to\infty} e^{-r/b} = \frac{q}{\varepsilon_0}$$

$q = 0$

5. The smallest angle of incoming beam should be greater than critical angle of the medium.



The smallest angle incoming beam should be greater than critical angle of the medium.

$$\sin\theta \geq \sin\theta_c$$

$$\frac{R}{R+a} \geq \frac{1}{\mu}$$

$$a \leq (\mu - 1)R$$

$$a_{max} = \frac{R}{2}$$

6. $$d\vec{B} = \frac{\mu_0 i}{4\pi}\,\frac{d\vec{\ell}\times\hat{r}}{r^2} = \frac{\mu_0 i}{4\pi r^2}\left(dr\hat{\ell}_r + rd\theta\hat{\ell}_\theta\right)\times\left(-\hat{\ell}_r\right) = \frac{\mu_0 i}{4\pi r^2} r\,d\theta\,\hat{k}$$

Where $\hat{\ell}_r$ and $\hat{\ell}_\theta$ are unit vectors along radial and azimuthal directions

$$d\vec{B} = \frac{\mu_0 i}{4\pi}\,\frac{d\theta}{r}\hat{k}$$

$$B = \int d\vec{B} = \int_0^\pi \frac{\mu_0 i}{4\pi}\,\frac{d\theta}{\left(a + \frac{b}{\pi}\theta\right)}\hat{k}$$

$$= \frac{\mu_0 i}{4b}\ln\left(1+\frac{b}{a}\right)\hat{k}$$

7. Since current leads the voltage
Hence circuit is capacitive.

8. For 1 mole gas,
$PV = RT$
$PV^2 = aR = \text{constant}$
Compare with $PV^x = \text{constant}$
$x = 2$
molar heat capacity of gas $C = \frac{R}{\gamma - 1} + \frac{-R}{x - 1}$

$$C = \frac{R}{\gamma - 1}(2 - \gamma)$$

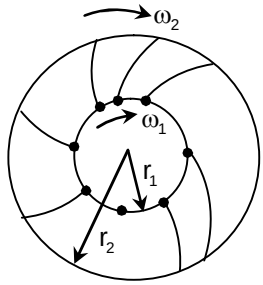
$$\Delta Q = nC\Delta T = \frac{R\Delta T}{\gamma - 1}(2 - \gamma)$$

9. As the sand leaves the inner drum through the open holes, it does not exert any force of the drum, angular momentum remains conserved.
At time t,

$$m_1 r_1^2 \omega_0 = (m_1 - \lambda t) r_1^2 \omega_0 + (m_2 + \lambda t) r_2^2 \omega_2$$

$$\omega_2 = \frac{\lambda t r_1^2 \omega_0}{(m_2 + \lambda t) r_2^2}$$

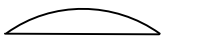
The speed of inner drum does not change.

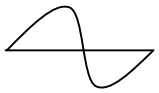


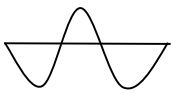
10. In first case, we can write torque equation about bottom point because it coincides with instantaneous axis of rotation while in second case, you will write torque equation of centre of mass.

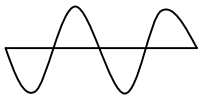
11. $V(A) \cong 5V$
$V(B) \cong V(C) \cong 2.5V$
$V(D) \cong 0$

12. $K = 0.5\pi = \frac{2\pi}{\lambda} \Rightarrow \lambda = 4\text{cm}$


$\Rightarrow L = \frac{\lambda}{2} = 2\text{cm}$


$\Rightarrow L = \lambda = 4\text{cm}$


$\Rightarrow L = \frac{3\lambda}{2} = 6\text{cm}$


$\Rightarrow L = 2\lambda = 8\text{cm}$

13. $\frac{\lambda}{2} = \ell = \frac{v}{2f} = \ell \cong 0.657\text{m}$

14. $f = \frac{nv}{2\ell}$

$n = 1$

$V = 344$

$\ell = 0.657 - 0.035$

$f = 276.5$ hz

15. $Hf > E_b$

$6.6\times10^{-34}\,\frac{\omega}{2\pi} > 10^{-3}\times1.6\times10^{-19}$

$\omega > 1.6\times10^{12}$ rad/s

16. $a = \frac{F_0}{m}\cos\omega t$

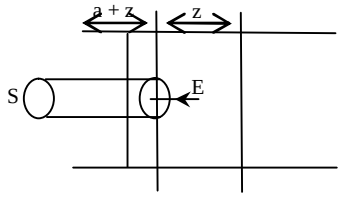
$v = \frac{-eF_0}{m\omega}\sin\omega t$

$|v_{max}| = \frac{eF_0}{m\omega}$

17. Here z is co-ordinate

$EA = \frac{-\rho_0(a+z)A}{\varepsilon_0}$

$E = \frac{-\rho_0(a+z)}{\varepsilon_0}$



18. $\int dv = -\int \vec{E}\cdot d\vec{r} = -\left[\int_a^0 \frac{\rho_0(a+z)}{\varepsilon_0}dz + \int_0^a \frac{-\rho_0(a-z)}{\varepsilon_0}dz\right]$

$= \frac{\varepsilon_0 a^2}{\varepsilon_0}$

SECTION – C

1. Initially,

$C = \frac{\varepsilon_0 A}{d}$ -----(1)

$V_0 = Ed$ ------(2)

and $V_f = Ed_f$

$4V_0 = E(d+x)$

$4Ed = E(d+x)$

$d = \frac{x}{3}$

Volume = A(d + x)

$= \frac{Cd}{\varepsilon_0}\cdot(d+x)$

$= \frac{4x^2C}{9\varepsilon_0}$

2. $V = \dfrac{\varepsilon}{1+\dfrac{R_1}{R_2}}$

$Q = CV = \dfrac{C\varepsilon}{1+\dfrac{R_1}{R_2}}$

3. When switch is open

$V_1 + V_2 = 6V$

$V_1 = V_2 = 3V$

When switch is closed

$V_1^1 = 2V; V_2^1 = 4V$

$\dfrac{V_2^1 - V_2}{V_1 - V_1^1} = \dfrac{4-3}{3-2} = 1$

4. If length of side = 2a = $60\left(\sqrt{2}+1\right)$ cm

angular momentum conservation,

$Mav_0 = I\,\omega$

$\omega = \dfrac{Mav_0}{I}$.....(1)

$\dfrac{1}{2}I\omega^2 \geq Mga\left(\sqrt{2}-1\right)$.....(2)

$I = I_{cm} + M\left(a\sqrt{2}\right)^2$

$= \dfrac{8Ma^2}{3}$....(3)

Solve these equation,

$v_0 \geq 4$ m/s

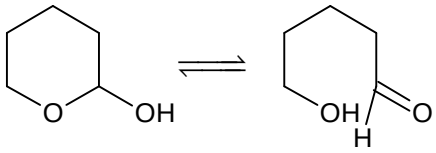
Chemistry

PART – II

SECTION – A

1. Tollen's reagent gives +ve test for $H-\overset{\overset{O}{\|}}{C}-OH$ not CH_3COOH.

2. Due to steric hindrance in II the loan pair e^- are not involved in resonance in I electron withdrawing effect of NO_2 is high.

3. 

4. $_{83}Bi^{211} \longrightarrow {}_{81}Tl^{207} + {}_2He^4$;
total time = n × half life
moles of substance left after n halve.
$= \frac{\text{initial moles}}{2^n} = \frac{2}{2^4} = 0.125$
moles of He produced
= 2 – 0.125 = 1.875
Pressure developed due to
$He = \frac{1.875 \times 0.0821 \times 300}{100}$
= 0.4618 atm

5. $K_{eq} = \frac{K_f}{K_b} = \frac{[CH_3]^2}{[C_2H_6]}$, $[CH_3] = 10^{-5}$m

$\frac{1.57 \times 10^{-3}}{K_b} = \frac{\left[10^{-5}\right]^2}{1}$, $K_b = 1.57 \times 10^7$ L mol^{-1}s^{-1}

6. $P_{H_2O} = 760 - 722 = \frac{38}{760}$atm;

$n_{H_2O} = \frac{0.9}{18}$

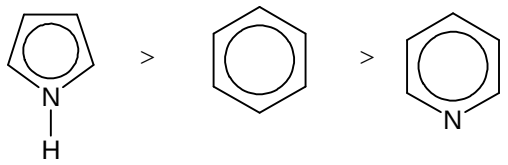
$V = \frac{0.9}{18} \times \frac{0.0821 \times 300}{38} \times 760$;

V = 24.63 L

7. KO_2 has only unpaired e^- remaining all are diamagnetic.

8. Both compounds resemble each other in solubility.

12. Order of reactivity towards E. A. substitution



13. From equation above $\Delta S = \dfrac{-d(\Delta G)}{dT}$

$$\Delta S = \frac{-d(-nFE)}{dT} = nF\frac{dE}{dT}(or)\left(\frac{dE}{dT}\right)_P = \frac{\Delta S}{nF}$$

14. $\Delta H = nF\left(\dfrac{dE}{dT}\right)_P T - nFE_{cell}$

$$-218 \times 1000 = 2 \times 96500 \times 300\left(\frac{dE}{dT}\right)_P - 2 \times 96500 \times 1.015$$

$$\left(\frac{dE}{dT}\right)_P = -3.81 \times 10^{-4}\, VK^{-1}$$

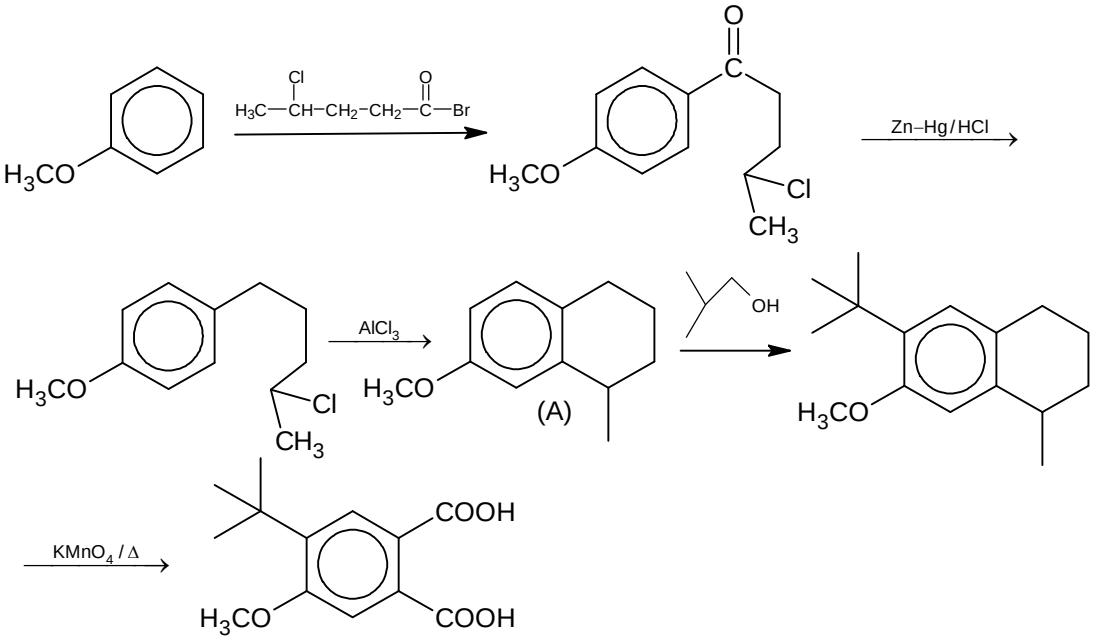
16. Mixture contains NaOH + $ZnSO_4$

$$NaOH + ZnSO_4 \longrightarrow \underset{\text{soluble}}{Na_2ZnO_2} + Na_2SO_4 + 2H_2O$$

$$Na_2ZnO_2 + 2HCl \longrightarrow 2NaCl + Zn(OH)_2 \downarrow$$

$$Zn(OH)_2 + 2HCl \longrightarrow \underset{\text{soluble}}{ZnCl_2} + 2H_2O$$

18.



SECTION –C

1. $\Delta E = \dfrac{hc}{\lambda} = \dfrac{(6.63 \times 10^{-34}\,J.s)(3 \times 10^{8}\,m/s)}{3.055 \times 10^{-8}\,m}$

$= 6.52 \times 10^{-18}$ J.

$\Delta E_H = \frac{3}{4}\left(2.176\times10^{-18}\,J\right)$

$= 1.63 \times 10^{-18}$

$\Delta E = \Delta E_H (Z)^2$

$Z^2 = \frac{\Delta E}{\Delta E_H} = \frac{\left(6.52\times10^{-18}\right)}{\left(1.63\times10^{-18}\right)} = 4$

$Z = 2$

2. $Sb_2S_3(s) + 3H_2(g) \rightleftharpoons 2Sb(s) + 3H_2S(g)$

$0.01 - x \quad 0.01 - 3x \quad 2x \quad 3x$

$3x = 0.005$

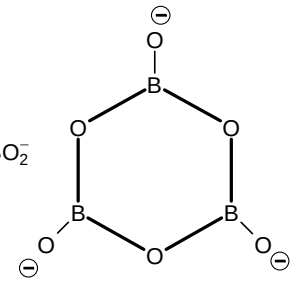
$H_2S + Pb^{2+} \longrightarrow PbS + 2H^+$

no. of moles of PbS formed = $\frac{1.19}{238} = 0.005$ moles

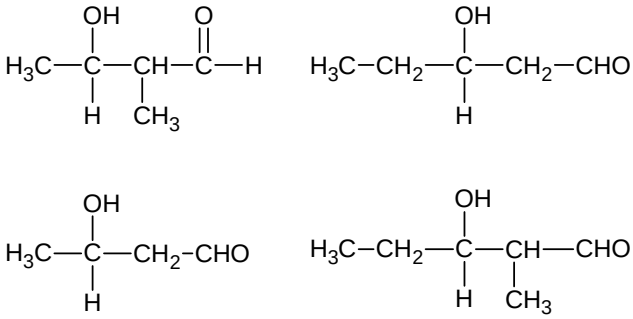
at equilibrium $\left[H_2\right] = \left[\frac{0.005}{250}\right]$

$K_c = \left[\frac{0.005}{0.005}\right]^3 = 1$

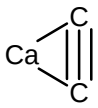
3. Metaborate BO_2^-



4.



5. $A = CaC_2$

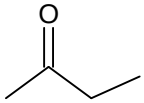


$B = HC\equiv CH$

$C = HC\equiv CNa$

$D = HC\equiv C-CH_2-CH_3$

$E =$



Mathematics

PART – III

SECTION – A

1. $a_1 = 1$

$a_2 = 1-\frac{1}{2}$

$a_3 = 1-\frac{1}{2}+\frac{1}{3}$

$\vdots$

$\vdots$

$a_\infty = 1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\ldots\ldots = \ln 2$

2. Given, $\sin\left(\frac{\pi}{2n}\right)+\cos\left(\frac{\pi}{2n}\right)=\frac{\sqrt{n}}{2}$

$\Rightarrow \sin^2\left(\frac{\pi}{2n}\right)+\cos^2\left(\frac{\pi}{2n}\right)+2\sin\left(\frac{\pi}{2n}\right)\cos\left(\frac{\pi}{2n}\right)=\frac{n}{4}$

$\Rightarrow 1+\sin\left(\frac{\pi}{n}\right)=\frac{n}{4} \Rightarrow \sin\left(\frac{\pi}{n}\right)=\frac{n-4}{4}$

For n = 2, the given equation is not satisfied considering n > 1 and $n \neq 2$, $0 < \sin\left(\frac{\pi}{n}\right) < 1$

$\Rightarrow 0 < \frac{n-4}{4} < 1 \Rightarrow 4 < n < 8$

3. $g(f(x)) = \tan\left(x-\frac{\pi}{4}\right)=\frac{\tan x-1}{\tan x+1}$

$\Rightarrow g(x)=\frac{x-1}{x+1}$

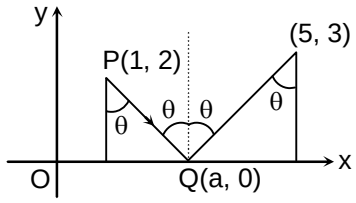
$f(g(x)) = \tan\left(\frac{x-1}{x+1}\right)$

4. Locus of point of intersection of perpendicular tangents is director circle for $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ equation of director circle is $x^2 + y^2 = a^2 – b^2$ which is real if a > b

5. Image of P in x-axis is (1, –2)

$\therefore$ Equation of reflected ray is $(y – 3) = \frac{5}{4}(x-5)$

$\Rightarrow 5x – 4y = 13$

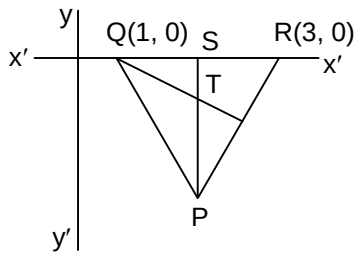


6. Radical centre of the circles described on the sides of a triangle as diameters is the orthocentre of the triangle

$\therefore$ OS = 2

$ST = -QS\tan\frac{\pi}{6} = -\frac{1}{\sqrt{3}}$

$\therefore$ Quadrants of T is $\left(2, -\frac{1}{\sqrt{3}}\right)$



7. $\therefore$ Since, (4, –4) and (9, 6) lie on $y^2 = 4a(x - b)$
$\therefore$ $4 = a(4 - b)$ and $9 = a(9 - b)$
$\therefore$ $a = 1$ and $b = 0$
$\therefore$ The parabola is $y^2 = 4x$
Let the points R be $(t^2, 2t)$
$\therefore$ Area of $\Delta PRQ = \frac{1}{2}\begin{vmatrix} 4 & -4 & 1 \\ 9 & 6 & 1 \\ t^2 & 2t & 1 \end{vmatrix} = -5\left(t - \frac{1}{2}\right)^2 + \frac{125}{4}$
$\therefore$ Area in largest when $t = \frac{1}{2}$
$\therefore$ Coordinate of the point R is $\left(\frac{1}{4}, 1\right)$

8. f(x) is a decreasing function and for major axis to be x-axis
$f(k^2 + 2k + 5) > f(k + 11)$
$\Rightarrow k^2 + 2k + 5 < k + 11$
$\Rightarrow k \in (-3, 2)$

9. $$\frac{x^{-7/6} - x^{5/6}}{x^{1/3}\left(x^2 + x + 1\right)^{1/2} - x^{1/2}\left(x^2 + x + 1\right)^{1/3}} = \frac{x^{-7/6} - x^{5/6}}{x^{5/6}\left(x + \frac{1}{x} + 1\right)^{1/2} - x^{5/6}\left(x + \frac{1}{x} + 1\right)^{1/3}}$$

$$= \frac{\frac{1}{x^2} - 1}{\left(x + \frac{1}{x} + 1\right)^{1/2} - \left(x + \frac{1}{x} + 1\right)^{1/3}} \text{ put } Z = \left(x + \frac{1}{x} + 1\right)^{1/6} - 1$$

After solving we get, P = 6, T = 2 and Q = 1

10. $\int e^{\tan^2 x} \sin 4x\,dx$
Applying integration by parts
$e^{\tan^2 x}\int \sin 4x\,dx - \int\left(e^{\tan^2 x} \times 2\tan x \sec^2 x \int \sin 4x\,dx\right)dx$
$= -\frac{1}{4}\cdot e^{\tan^2 x}\cdot \cos 4x + \frac{1}{2}\int e^{\tan^2 x}\tan x \sec^2 x\left[2\left\{2\cos^2 x - 1\right\}^2 - 1\right]dx$ continue solving like this

11. Put $z = \cos x$, $dz = -\sin x\,dx$
$$f(t) = \int_{-\frac{\pi}{2}}^{t} \frac{\sin^3 x}{\sqrt[3]{\cos^2 x}}dx = -\int_{-\frac{\pi}{2}}^{t}\left(1 - z^2\right)z^{-\frac{2}{3}}\,dz$$

12. If a, b and c are odd integers then discriminant = $b^2 - 4ac \neq$ perfect square
$\Rightarrow$ Roots are always irrational
If $ac = 1 \Rightarrow a = 1$, $c = 1$ and $b \geq 3$
$\because$ a, b and c = odd
$\therefore$ $c(a - b + c) < 0 \Rightarrow f(0)\, f(-1) < 0$
$\Rightarrow \alpha \in (-1, 0)$

13.-14. Total points generated = 20
Two points can be selected in ${}^{20}C_2 = 190$ ways
(i) Number of points with all coordinates odd = 4
(ii) Number of points with x odd, y even = 6

(iii) Number of points with x even, y odd = 4
(iv) Number of points with both even = 6
$\therefore$ Required number of points = ${}^4C_2 + {}^6C_2 + {}^4C_2 + {}^6C_2 = 42$

(1) Required probability = $\frac{42}{190} = \frac{21}{95}$

(2) $(x_1 \neq x_2$ and $y_1 = y_2) = 2 \times 5 = 10$

Required probability = $\frac{10}{190} = \frac{1}{19}$

15. $\sin x + \cos x = 1 + \sin x \cos x$

$\Rightarrow t = 1 + \frac{t^2 - 1}{2}$ (Let $\sin x + \cos x = t$)

$2t = 2 + t^2 - 1 \Rightarrow t^2 - 2t + 1 = 0$

$(t - 1)^2 = 0 \Rightarrow t = 1$

$\therefore \sin x + \cos x = 1$

$\cos\left(x - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

$x - \frac{\pi}{4} = 2n\pi \pm \frac{\pi}{4}$

$x = 2n\pi \pm \frac{\pi}{4}$

$\Rightarrow x = 2n\pi$ or $x = 2n\pi + \frac{\pi}{2}$, $n \in I$

16. $\sin^4 x + \cos^4 x = \sin x \cos x$

$(\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x = \sin x \cos x$

$1 - 2p^2 = p$ ($\because \sin x \cos x = p$)

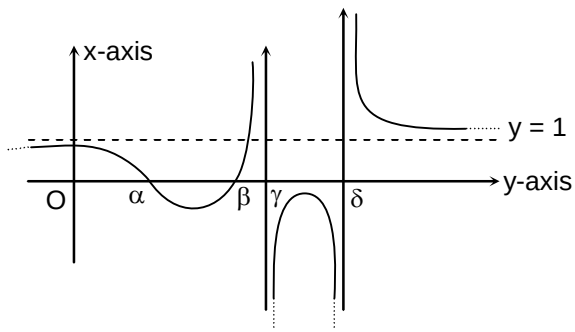
$(p + 1)(2p - 1) = 0$

$\sin x \cos x = -1$ $\quad 2 \sin x \cos x - 1 = 0$

$\sin x \cos x = -1$ $\quad \sin 2x = 1$

$\sin 2x = -2$ (not possible) $\quad 2x = 2n\pi + \frac{\pi}{2} \Rightarrow x = (4n + 1)\frac{\pi}{4}$

17.-18. Graph of $f(x) = \frac{(x - \alpha)(x - \beta)}{(x - \gamma)(x - \delta)}$



SECTION – C

1. Let $g(x) = (x + 1)\, p(x) - 1$ ..... (1)

$g(0) = g(1) = g(2) = ... = g(11) = 0$

$\therefore g(x) = kx(x - 1)(x - 2) ..... (x - 11)$ ..... (2)

Also, $g(-1) = -1$

$\therefore k = -\frac{1}{\lfloor 12}$

Now, from equation (1) and (2), we get

$(x+1)p(x)-1=-\frac{1}{\underline{|12}}x(x-1)(x-2).....(x-11)$

$\therefore$ p(12) = 0

2. ${}^nC_1, {}^nC_2, {}^nC_3$ an in A.P.

$\frac{2n(n-1)}{2}=n+\frac{n(n-1)(n-2)}{6}$

$\Rightarrow$ n = 2 or 7

Since ${}^nC_3$ demands that n should be ≥ 3 then n can not be 2

$\therefore$ n = 7

3. $x_0 = 2 \Rightarrow y_0 = \frac{8}{4} = 2$

$\therefore$ Equation of the tangent is $y = -2x + 6$ which cuts x-axis at (3, 0)

Hence B $\equiv$ (3, 0)

$\therefore$ $\overrightarrow{AB} = (3-2)\hat{i} + (0-2)\hat{j}$ and $\overrightarrow{OB} = 3\hat{i}$

$\therefore$ $\overrightarrow{AB} \cdot \overrightarrow{OB} = 3$

4. $f'(x) = -2 - 3x^2 < 0 \Rightarrow f(x)$ is decreasing

$\because$ f(f(f(x))) > f(f(–x))

$\Rightarrow$ f(f(x)) < f(–x) $\Rightarrow$ f(x) > –x

$\Rightarrow 30 - 2x + x^3 > -x \Rightarrow x^3 + x - 30 < 0$

$\Rightarrow (x-3)(x^2 + 3x + 10) < 0 \Rightarrow x < 3$

Positive values of integer x = 1, 2

5. $\begin{matrix} x+\sin y = 2010 \\ x+2010\cos y = 2009 \\ \sin y - 2010\cos y = 1 \end{matrix}$ substracting

$\Rightarrow$ sin y = 1 + 2010 cos y

Possible if $y = \frac{\pi}{2}$ $\because$ $0 \leq y \leq \frac{\pi}{2}$

$\therefore$ x = 2010 – 1 = 2009

$\therefore$ $\frac{[x+y]}{1005} = \frac{\left[2009+\frac{\pi}{2}\right]}{1005} = 2$